Global identifiability of a behavioral SIR model for every integer saturation exponent $k \geq 2$

Tatsuhiko Takimoto

Independent researcher

Version 1.1 (draft), 22 October 2026

Abstract

Pant, Saucedo and Pogudin studied a behavioral SIR model in which the transmission rate is divided by $(1+\alpha I)^k$, and conjectured that for $k \geq 2$ it is globally structurally identifiable from continuous, noise-free observation of the number of infectious individuals $I(t)$, with the total initial population known. We prove this for every fixed integer $k \geq 2$: generically, the parameters α, β, γ and the initial values of S, I and R are uniquely determined by $I(t)$. The proof has two parts. First, the minimal input–output equation is written in closed form for all k; its coefficients determine α, γ and β through rational formulas whose denominators are k and $k(k-1)\alpha^2$, so no integer $k \geq 2$ is exceptional. Second, a valuation argument shows that the two-state model obtained by eliminating R has no nonconstant rational first integral; the argument uses only that $k \geq 2$. Results of Ovchinnikov, Pogudin and Thompson then convert the coefficient statement into global identifiability. The proof is written in prose and is not formalized in a proof assistant. Exact computer-algebra checks for $k = 1, \ldots, 12$, and a separately written check for $k = 2, \ldots, 9$, agree with the general formulas; they are finite checks and do not replace the proof. Non-integer exponents, an unknown exponent, and estimation from discrete or noisy data are outside the scope of the result.

1 Introduction

Pant, Saucedo and Pogudin examined which epidemiological quantities are structurally identifiable in a range of compartmental models [1]. In their Section 3.6 they consider a behavioral extension of the SIR model in which contacts decrease as infections increase,

$$S' = -\frac{\beta SI}{(1+\alpha I)^k}, \qquad I' = \frac{\beta SI}{(1+\alpha I)^k} - \gamma I, \qquad R' = \gamma I, \tag{1}$$

observed through $y = I$, with the total initial population $N_0 = S(0) + I(0) + R(0)$ known. For $k = 1$ they proved that α, $\beta\gamma$ and $\alpha\gamma + \beta$ are globally identifiable while β and γ are only locally identifiable, the two-fold ambiguity being $(\beta, \gamma) \mapsto (\alpha\gamma, \beta/\alpha)$ [1, Theorem 3.8], [2, Section S8]. Using StructuralIdentifiability.jl, they found global identifiability for $k = 2, 3, 4, 5, 10$ and 50 [2, Section S9], and they stated global identifiability for $k \geq 2$ as Conjecture 2 of [1]. Their Remark 1 in [2, Section S9] sketches why the property should hold for all sufficiently large k: after the substitution $A = (1+\alpha I)^k/\beta$, the parameters and initial conditions are rational functions of the output jet, whose denominators can vanish for at most finitely many values of k; the exceptional values are not determined.

The tools we use are due to Ovchinnikov, Pogudin and Thompson [4]. They showed that the coefficients of a monic characteristic presentation of the input–output ideal generate the field of input–output identifiable functions [4, Corollary 5.7], and that input–output identifiability coincides with identifiability when the model has no rational first integrals [4, Theorem 4.7]. The second hypothesis is essential here: with all three states, $S + I + R$ is a rational first integral, so we work with the two-state model on the surface of known N_0.

Result. Theorem 6 proves global identifiability for every fixed integer $k \geq 2$. Its new ingredients are a closed form of the minimal input–output equation valid for all k, an explicit rational inverse of its coefficient map (Proposition 4), and a lemma that the reduced model has no nonconstant rational first

integral (Lemma 5). The hypothesis $k \geq 2$ is used essentially in these two places, each time through a single, visible fact: a coefficient $k(k-1)\alpha^2\gamma$ that vanishes for $k = 1$, and a pole of order k that cannot be a logarithmic derivative when $k \geq 2$. Both facts are consistent with the $k = 1$ non-identifiability result of [1].

2 What is proved, and how

Table 1 lists the claims of this article and their present basis. All proofs are written in prose; none is formalized in a proof assistant, and the article has not undergone human expert review. Who ran which check, and when, is recorded in the verification record that accompanies the article (`VERIFICATION-RECORD.md`).

Claim	Basis	Range
Input–output equation (4), its irreducibility and characteristic presentation (Section 4)	Elementary algebra (Gauss's lemma) and the definitions of [4]	Every integer $k \geq 2$
Coefficient table and inverse map (Proposition 4)	Binomial theorem	Every integer $k \geq 2$
No rational first integral (Lemma 5)	Formal Laurent expansion and logarithmic derivatives	Every integer $k \geq 2$
Global identifiability (Theorem 6)	The above together with [4, Theorem 4.7, Corollary 5.7, Proposition 3.1]	Every integer $k \geq 2$, generic parameters and initial values
Coefficients, vector-field identity, recovery formulas	Exact computer algebra, two separately written implementations (Section 8)	$k = 1, \ldots, 12$ and $k = 2, \ldots, 9$
Uniqueness for one trajectory	Exact power series and linear algebra	$k = 2, 3, 5$, one rational point
Not claimed	Non-integer k; unknown k; an explicit description of the exceptional set; practical identifiability from finite or noisy data	

Table 1: Claims and their basis. The finite computations in the lower part are checks of the general formulas and do not replace the proof.

The proof depends on three results of [4], whose hypotheses we verify explicitly (Proposition 3 and Lemma 5). Everything else is ordinary commutative algebra.

3 Model and identifiability

Throughout, $k \geq 2$ is a fixed integer, known to the observer. The parameters are $\mu = (\alpha, \beta, \gamma)$, and we write $K = \mathbb{C}(\alpha, \beta, \gamma)$ for the field of rational functions in them. We write initial values as $S(0)$, $I(0)$, $R(0)$.

Reduction to two states. The equations for S and I in (1) do not contain R. If (S, I, R) solves (1) with $S + I + R = N_0$, then (S, I) solves

$$S' = -\frac{\beta S I}{u^k}, \qquad I' = \frac{\beta S I}{u^k} - \gamma I, \qquad u = 1 + \alpha I, \qquad y = I, \tag{2}$$

and conversely a solution of (2) together with $R = N_0 - S - I$ solves (1), since $(N_0 - S - I)' = \gamma I$. Both have the same output. Hence $\alpha, \beta, \gamma, S(0), I(0), R(0)$ are identifiable for (1) with known N_0 exactly when $\alpha, \beta, \gamma, S(0), I(0)$ are identifiable for (2), and then $R(0) = N_0 - S(0) - I(0)$. The generic sets correspond under projection to $(\mu, S(0), I(0))$. Pant, Saucedo and Pogudin use the same reduction [2, (S9.2)], as does the authors' published code for $k = 2$ [3]. Only the last step uses N_0.

Identifiability. We use the definition of [4, Definition 2.2], specialized to a model without inputs. Write (2) as $x' = f/Q$, $y = g/Q$ with $x = (S, I)$, $Q = u^k$, $f = (-\beta SI,\ \beta SI - \gamma I u^k)$ and $g = Iu^k$. For $(\hat{x}_0, \hat{\mu}) \in \mathbb{C}^2 \times \mathbb{C}^3$ with $Q(\hat{x}_0, \hat{\mu}) \neq 0$, let $Y(\hat{x}_0, \hat{\mu})$ be the output of the unique solution that is analytic near $t = 0$.

Definition 1. A rational function $h(x_0, \mu)$ is *identifiable* if there is a nonempty Zariski open set $\Theta \subset \mathbb{C}^5$ such that, for every $(\hat{x}_0, \hat{\mu}) \in \Theta$ at which h and Q^{-1} are defined, every $(\tilde{x}_0, \tilde{\mu})$ at which h and Q^{-1} are defined and with $Y(\tilde{x}_0, \tilde{\mu}) = Y(\hat{x}_0, \hat{\mu})$ satisfies $h(\tilde{x}_0, \tilde{\mu}) = h(\hat{x}_0, \hat{\mu})$. The model is *globally identifiable* if each parameter and each component of x_0 is identifiable.

Pant, Saucedo and Pogudin work over the reals with an open dense set of admissible points [1, Definition 1]. As they note [1, Remark 4], identifiability over $\mathbb{C}$ implies identifiability over $\mathbb{R}$: the real points of a nonempty Zariski open subset of $\mathbb{C}^5$ form an open dense subset of $\mathbb{R}^5$, and a real competing point is also a complex one. The set of real points of Θ, intersected with the region of positive parameters and populations, is open and dense in that region. Identifiability is a generic statement. For example, $I(0) = 0$ gives the zero output, and $S(0) = 0$ gives pure exponential decay, from which α and β cannot be recovered.

4 The input–output equation

Let D be the derivation of $L = K(S, I)$ defined by (2), with D trivial on K; here S and I are independent indeterminates, and α, β, γ, being indeterminates, are nonzero in K. This differential field is the field $\mathrm{Frac}(\mathbb{C}(\mu)\{x, y\}/I_\Sigma)$ of [4]: with $I_\Sigma = [Qx' - f,\ Qy - g] : Q^\infty$ [4, Notation 2.4], the ideal I_Σ is prime and consists of the differential polynomials that vanish on all solutions [5, Lemma 3.2]; see also the proof of [4, Corollary 5.7]. It is the kernel of the differential homomorphism $\mathbb{C}(\mu)\{x, y\} \to L$ sending x to (S, I) and y to I: every differential polynomial is congruent modulo I_Σ, after multiplication by a power of Q, to a polynomial in S, I. If the original element maps to zero in L, that polynomial is zero, because S and I are algebraically independent over K; since I_Σ is saturated with respect to Q, the element lies in I_Σ.

Put $y = I$, $v = Dy$ and $w = D^2 y$. The second equation of (2) and $D(S + I) = -\gamma I$ give

$$S = \frac{(v + \gamma y)\, u^k}{\beta y}, \qquad DS = -(v + \gamma y). \tag{3}$$

Differentiating the first identity, substituting the second and multiplying by βy^2, we obtain $P(y, v, w) = 0$ on the model, where

$$P = yu^k w - \big(1 - (k-1)\alpha y\big)u^{k-1}v^2 + \big(k\alpha\gamma y^2 u^{k-1} + \beta y^2\big)v + \beta\gamma y^3. \tag{4}$$

Indeed, the derivative of u^k contributes $k\alpha u^{k-1}v$; the v^2 terms combine through $k\alpha y - u = -(1-(k-1)\alpha y)$, and the two terms $\pm\gamma y u^k v$ cancel, leaving $k\alpha\gamma y^2 u^{k-1}v$.

Lemma 2. *The elements y and v of L are algebraically independent over K, and P is irreducible in $K[y, v, w]$.*

Proof. The map $(S, I) \mapsto (y, v)$ with $v = \beta Sy/u^k - \gamma y$ is inverted by (3), so $K(S, I) = K(y, v)$, and y, v are algebraically independent. Write $P = Aw + B$ with $A = yu^k$ and $B \in K[y, v]$. The only irreducible factors of A in $K[y]$ are y and $u = 1 + \alpha y$. From (4),

$$B(0, v) = -v^2, \qquad B(-1/\alpha, v) = \frac{\beta}{\alpha^2}\Big(v - \frac{\gamma}{\alpha}\Big),$$

where the second identity uses $k \geq 2$, so that every term containing u^{k-1} vanishes at $u = 0$. (For $k = 1$ the value is also nonzero and the lemma still holds; we restrict to $k \geq 2$ only for brevity.) Both are nonzero, so neither y nor u divides B, and $\gcd(A, B) = 1$. Thus P is primitive and of degree one in w over $K[y, v]$; it is irreducible over $K(y, v)$ and hence, by Gauss's lemma, in $K[y, v, w]$. □

Proposition 3. *Let $J = I_\Sigma \cap K\{y\}$, the kernel of the differential homomorphism $K\{y\} \to L$, $y \mapsto I$. For the ranking $y < y' < y'' < \cdots$, the set $\{P\}$, with y', y'' in place of v, w, is a monic characteristic presentation of J in the sense of [4, Definition 5.4].*

Proof. J is prime, being the kernel of a map to a field, and $P \in J$. In one differential indeterminate an autoreduced set has at most one element. The leader of P is y'' and has degree one, so an element of J of lower rank would lie in $K[y, y']$; by Lemma 2 there is no nonzero such element. Hence $\{P\}$ is a characteristic set of J [4, Definition 5.3]. Its non-leading variables are y, y'. The initial $A = yu^k$ lies in $K[y] \subset K[y, y']$, and, P being irreducible and of positive degree in y'', P has no factor in $K[y, y'] \setminus K$. The coefficient of yy'' equals 1, so $\{P\}$ is monic in the sense of [4, Definition 5.4]. □

By [4, Corollary 5.7], the field of input–output identifiable functions is generated over $\mathbb{C}$ by the coefficients of P, as a polynomial in y, y', y''.

5 Coefficients and their inversion

For a monomial m in y, v, w write $[m]P$ for its coefficient in P, and set $\binom{n}{r} = 0$ outside $0 \le r \le n$.

Proposition 4. *The coefficients of P, including those equal to zero, are*

$$\begin{aligned} [y^{j+1}w]P &= \binom{k}{j}\alpha^j, && 0 \le j \le k, \\ [y^{j+1}v]P &= k\binom{k-1}{j-1}\gamma\alpha^j + \delta_{j1}\beta, && 1 \le j \le k, \\ [y^j v^2]P &= -\Big(\binom{k-1}{j} - (k-1)\binom{k-1}{j-1}\Big)\alpha^j, && 0 \le j \le k, \\ [y^3]P &= \beta\gamma, \end{aligned} \tag{5}$$

and all other monomials have coefficient zero. In particular

$$[yw]P = 1, \quad [y^2w]P = k\alpha, \quad [y^2v]P = \beta + k\alpha\gamma, \quad [y^3v]P = k(k-1)\alpha^2\gamma, \tag{6}$$

and, for $\alpha\gamma \neq 0$,

$$\alpha = \frac{[y^2w]P}{k}, \qquad \gamma = \frac{[y^3v]P}{k(k-1)\alpha^2}, \qquad \beta = [y^2v]P - k\alpha\gamma. \tag{7}$$

Proof. Expand u^k and u^{k-1} in (4) by the binomial theorem. The v^2 coefficient is $-(u^{k-1} - (k-1)\alpha y\, u^{k-1})$, which gives the third line; the second line comes from $k\alpha\gamma y^2 u^{k-1}$ and βy^2. Equations (6) are the cases $j = 0, 1$ of the first line and $j = 1, 2$ of the second, and (7) solves them successively. □

For instance, $[yv^2]P = 0$ for every k. The denominators in (7) are k and $k(k-1)\alpha^2$, which are nonzero in K for every integer $k \ge 2$. Since (6) lies in the field generated by the coefficients and α, β, γ are rational in (6), Proposition 3 and [4, Corollary 5.7] show that α, β and γ are input–output identifiable, and that the field of input–output identifiable functions is all of K.

For $k = 1$ the coefficient $[y^3v]P$ vanishes, the field generated by (5) is $\mathbb{C}(\alpha, \beta + \alpha\gamma, \beta\gamma)$, and the substitution $(\beta, \gamma) \mapsto (\alpha\gamma, \beta/\alpha)$ fixes all coefficients. This agrees with [1, Theorem 3.8]; formulas (7) must not be applied for $k = 1$.

6 No rational first integrals

Input–output identifiability does not by itself imply identifiability [4, Example 4.8]. The bridge is the following lemma.

Lemma 5. *Let $k \ge 2$. The constants of the differential field (L, D), $L = K(S, I)$, are exactly K.*

Proof. Use the coordinates S and $u = 1 + \alpha I$, so that $L = K(S, u)$, and the derivation $E = (u^k/I)\, D$. Since u^k/I is a nonzero element of L, E and D have the same constants. By (2),

$$ES = -\beta S, \qquad Eu = \alpha\beta S - \alpha\gamma u^k, \qquad E = -\beta S\, \partial_S + (\alpha\beta S - \alpha\gamma u^k)\, \partial_u. \tag{8}$$

Let $0 \neq F \in K(S,u)$ with $EF = 0$. Embed $K(u)(S)$ into the field of formal Laurent series $K(u)((S))$; no convergence is needed. Write

$$F = \sum_{n \geq m} f_n(u)\, S^n, \qquad f_m \neq 0.$$

Because E does not lower the S-order, it extends termwise to $K(u)((S))$ compatibly with the embedding. The term $\alpha\beta S\,\partial_u$ raises the order by one, so the coefficient of S^m in EF is

$$-\beta m f_m - \alpha\gamma u^k f_m' = 0. \tag{9}$$

Suppose $m \neq 0$. Then $f_m' \neq 0$, and

$$\frac{f_m'}{f_m} = -\frac{\beta m}{\alpha\gamma}\, u^{-k}.$$

Write $f_m = u^r h$ with $r \in \mathbb{Z}$ and $h \in K(u)$ having u-adic valuation zero. Then $f_m'/f_m = r/u + h'/h$ has u-adic valuation at least -1, while the right-hand side has valuation $-k \leq -2$. This contradiction shows $m = 0$. Then (9) gives $f_0' = 0$, so $f_0 \in K$ because the characteristic is zero. Now $F - f_0$ is again a constant; if it were nonzero, its S-order would be a positive integer, which the same argument excludes. Hence $F = f_0 \in K$. □

The order of the pole, k, is the only place where the exponent enters. For $k = 1$ the right-hand side has a simple pole and the argument fails. The lemma concerns the two-state model: with three states, $S + I + R$ is a nonconstant rational first integral, and [4, Theorem 4.7] cannot be applied directly.

7 Proof of the main theorem

Theorem 6. *Let $k \geq 2$ be a fixed integer. For model (1) with output $y = I$ and known $N_0 = S(0) + I(0) + R(0)$, the parameters α, β, γ and the initial values $S(0), I(0), R(0)$ are globally identifiable in the sense of Definition 1. The parameters and $S(0), I(0)$ are identifiable without using N_0. Explicitly, α, β, γ are given by (7) from the normalized minimal input–output equation, and*

$$S(t) = \frac{\big(y'(t) + \gamma y(t)\big)\big(1 + \alpha y(t)\big)^k}{\beta y(t)}, \qquad I(t) = y(t), \qquad R(t) = N_0 - S(t) - I(t) \tag{10}$$

near $t = 0$, whenever $\beta I(0)(1 + \alpha I(0)) \neq 0$.

Proof. By the reduction in Section 3 it suffices to treat the two-state model (2). By Lemma 5 the model has no rational first integral, that is, the constants of $\mathrm{Frac}(\mathbb{C}(\mu)\{x, y\}/I_\Sigma) = L$ are $\mathbb{C}(\mu)$. By [4, Theorem 4.7], a function of the parameters is identifiable if and only if it is input–output identifiable. Section 5 shows that α, β, γ are input–output identifiable; hence they are identifiable.

For the initial values we use the criterion [4, Proposition 3.1]: a rational function of (x_0, μ) is identifiable if and only if its image in L, with $S(0), I(0)$ replaced by S, I, lies in the subfield M generated by the image of $\mathbb{C}\{y\}$, that is, by $I, DI, D^2I, \ldots$. Applied to α, β, γ the criterion gives $\alpha, \beta, \gamma \in M$. By (3), $S = (DI + \gamma I)(1 + \alpha I)^k/(\beta I) \in M$, and $I \in M$. Therefore $S(0)$ and $I(0)$ are identifiable. Finally $R(0) = N_0 - S(0) - I(0)$.

Formula (10) is (3) evaluated on a solution. The condition $\beta I(0)(1 + \alpha I(0)) \neq 0$ defines a nonempty Zariski open set; on it y and $1 + \alpha y$ do not vanish near $t = 0$, so the right-hand side is defined there. Alternatively, analytically: once the parameters of any competing point agree with the true ones on the generic set, the same output gives the same right-hand side in (10), hence the same states. □

The theorem asserts uniqueness for generic, not for all, parameters and initial values. We do not determine the smallest exceptional set. In particular we do not claim that every point of an exceptional set is non-identifiable.

8 Finite exact checks

The general argument above does not rely on computation. The following exact computations were carried out as checks of the formulas; they are finite and do not replace the proof.

1. *Elimination from the vector field, $k = 1, \dots, 12$.* In a Python 3 program using only the standard library (sparse polynomials with integer coefficients and exact rationals), S is eliminated from the relation that is linear in S, $\beta S y - (v + \gamma y)u^k = 0$, and its total derivative by a 2×2 resultant. The result agrees with (5) in every monomial. Substituting the vector field into P and clearing denominators gives the zero polynomial, as an identity rather than at sample points. The four coefficients (6), the values $B(0, v)$ and $B(-1/\alpha, v)$ from Lemma 2, and the coordinate form (8) of the derivation were also checked. For $k = 1$ the parameter points $(\alpha, \beta, \gamma) = (2, 3, 5)$ and $(2, 10, 3/2)$, related by the substitution of Section 5, give identical coefficients; for $k = 2, \dots, 12$ they do not.

2. *Comparison with published equations.* The input–output equations printed in [2, Section S9] for $k = 2, 3, 4$, obtained there with StructuralIdentifiability.jl, coincide term by term with $-P$.

3. *A separately written implementation.* A second check, written without reference to the first and using the computer algebra system SymPy 1.13.3, verifies (4) on the model for $k = 2, 3, 5$ and for symbolic k (treating $(1 + \alpha I)^k$ as a separate symbol with the induced derivative), the coefficient table (5) and the inverse (7) for $k = 2, \dots, 9$, irreducibility of P over $\mathbb{Q}(\alpha, \beta, \gamma)$ for $k = 2, 3, 5$, and the leading-order identity (9). For $k = 2, 3$ and random rational parameters it finds no nonconstant polynomial first integral of degree at most 6 in (S, u); this is consistent with Lemma 5 and is not a proof of it.

4. *One trajectory.* For $k = 2, 3, 5$ and $(\alpha, \beta, \gamma, S(0), I(0)) = (1/3, 2/5, 1/7, 5/2, 1/2)$, the solution of (2) was computed as an exact rational power series to order 40. Among the polynomials $\sum c_{ijl}\, y^i (y')^j (y'')^l$ with $i \le k + 1$ and $j + 2l \le 2$, those vanishing on this trajectory form a one-dimensional space; normalizing $c_{101} = 1$ and applying (7) returns the original parameters exactly. Since the input–output equation of any competing parameter point lies in the same space of polynomials and has the same normalized coefficient, this computation shows directly that this particular trajectory determines the parameters.

Scripts for these checks accompany this article, and the verification record lists their outputs. Each check runs in a few seconds.

9 Scope and remarks

Integer exponents. Conjecture 2 of [1] is stated for $k \ge 2$, and the model text calls $k \ge 1$ the saturation exponent without restricting it to integers. Theorem 6 covers every fixed integer $k \ge 2$. For non-integer k the right-hand side of (1) is not rational, the framework of [4] does not apply as stated, and we make no claim. We also do not treat k as an unknown parameter to be recovered from the data.

Relation to the large-k sketch. Remark 1 of [2, Section S9] obtains global identifiability for all k outside an unspecified finite set from the software result for the augmented model. Theorem 6 shows that no integer $k \ge 2$ is an exception to global identifiability, and Proposition 4 gives the recovery formulas explicitly.

What is not addressed. Structural identifiability concerns exact, continuous observation. The theorem says nothing about the stability of estimating α, β, γ from finitely many noisy measurements, about the fit of (1) to real epidemics, or about identifiability at the non-generic parameter values excluded above.

Prior work. A literature search limited to the preprint versions of [1], the authors' code and keyword searches found no other proof for general k; it was not exhaustive, and we do not claim priority beyond it. The argument is short once the results of [4] are available, and it may be known to specialists in a different form.

AI-use statement

AI systems assisted with the derivation and checking of the proof, the exact computer-algebra checks, independent readings of the proof and of the manuscript, literature organization and manuscript drafting. The author is responsible for the mathematical claims and their stated scope; AI systems are not authors.

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